

Knowledge Test

Question 01

Multiple choice · Conceptual taxonomy

Which of the following statements correctly describes the difference between IBNYR and IBNER in the context of technical reserves?

- A. IBNYR refers to claims already reported whose individual case reserves are insufficient; IBNER refers to claims that have not yet been reported to the insurer.
- B. IBNYR refers to claims that have occurred but have not yet been reported to the insurer; IBNER refers to claims already reported whose future development (deterioration, litigation, reopening) will render the current reserve insufficient.
- C. Both terms are interchangeable synonyms; the distinction is merely one of Anglo-Saxon versus continental European terminology.
- D. IBNYR applies exclusively to life insurance lines with a savings component; IBNER is reserved for non-life lines with long-tail claims.

Answer

Option B

Standard actuarial taxonomy breaks down **IBNR in the broad sense** into two distinct components:

IBNYR (Incurred But Not Yet Reported): claims that occurred during the coverage period but have not been reported to the insurer as of the valuation date. No claim file has been opened. They are estimated using reporting frequency triangles.

IBNER (Incurred But Not Enough Reserved): claims already reported with active files, but whose individual case reserves will become insufficient due to future development — deterioration

of the damage, initiation of litigation, or late medical assessments. They are estimated using development triangles for reported claims.

The distinction is operationally relevant because it affects the estimation method, the required data sources, and the allocation of reserve shortfalls between claims management and actuarial departments.

Question 02

Given the cumulative paid claims triangle (COP millions), apply the *Chain-Ladder* method with volume-weighted development factors and calculate the total IBNR reserve.

Accident year	Development 1	Development 2	Development 3 (ult.)
2022	800	1,100	1,240
2023	920	1,265	—
2024	1,050	—	—

Assume a tail factor of 1.0 (development period 3 is the final period). Show the calculation of each factor and the projected ultimate claims by year.

Answer

Step 1 — Volume-weighted development factors

$$f_{1 \rightarrow 2} = \frac{1,100 + 1,265}{800 + 920} = \frac{2,365}{1,720} = 1.3750,$$

$$f_{2 \rightarrow 3} = \frac{1,240}{1,100} \approx 1.1273.$$

The tail factor is 1.0. Unrounded factors are used in the following calculations.

Step 2 — Cumulative development factors (CDFs)

From each period to ultimate:

$$\text{CDF}_{1 \rightarrow \text{ult}} = f_{1 \rightarrow 2} f_{2 \rightarrow 3} = 1.3750 \times \frac{1,240}{1,100} = 1.5500,$$

$$\text{CDF}_{2 \rightarrow \text{ult}} = f_{2 \rightarrow 3} \approx 1.1273,$$

$$\text{CDF}_{3 \rightarrow \text{ult}} = 1.0000 \quad (\text{fully developed}).$$

Step 3 — Completed triangle and IBNR by year

The projected values complete the cumulative paid claims triangle (COP millions):

Accident year	Development 1	Development 2	Development 3 (ult.)
2022	800	1,100	1,240
2023	920	1,265	1,426
2024	1,050	1,443.75	1,627.5

The 2022 accident year is already fully developed. For the remaining years:

$$\text{Ultimate}_{2023} = 1,265 \times \frac{1,240}{1,100} = 1,426.0,$$

$$\text{Ultimate}_{2024} = 1,050 \times 1.5500 = 1,627.5.$$

The reserve is obtained by subtracting the cumulative paid amount observed on the latest diagonal from the ultimate:

$$\text{IBNR}_{\text{year}} = \text{Ultimate}_{\text{year}} - \text{Diagonal}_{\text{year}}.$$

Year	Ultimate	Diagonal	IBNR
2022	1,240.0	1,240.0	0.0
2023	1,426.0	1,265.0	161.0
2024	1,627.5	1,050.0	577.5
Total			738.5

$$\text{IBNR}_{\text{total}} = 161.0 + 577.5 = 738.5 \text{ COP million}$$

Weighted averages versus simple averages

Volume-weighted averages (rather than simple averages) are used so that years with larger portfolios receive greater weight in the estimated factor — standard practice when the portfolio is growing. Using simple averages:

$$f_{1 \rightarrow 2}^{\text{simple}} = \frac{1.375 + 1.375}{2} = 1.375,$$

$$f_{2 \rightarrow 3}^{\text{simple}} = \frac{1,240/1,100}{1} \approx 1.1273.$$

Both methods coincide in this case: the two individual $1 \rightarrow 2$ factors are equal, and only one observation is available for $2 \rightarrow 3$. A difference may arise with more years in the triangle, when individual factors and volumes differ across years.

Question 03

Bornhuetter-Ferguson

For accident year 2024 in the preceding triangle, apply the *Bornhuetter-Ferguson* method with an a priori ultimate claims estimate of COP 1,580 million. Calculate the BF IBNR reserve and compare it with Chain-Ladder.

Use the CDFs obtained in Question 02. Show the decomposition into the observed component and the a priori component.

Answer

Bornhuetter-Ferguson formula

$$\text{Ultimate}_{\text{BF}} = \text{Paid} + \text{A priori ultimate} \times \left(1 - \frac{1}{\text{CDF}}\right).$$

For 2024:

$$\begin{aligned}\text{CDF}_{1 \rightarrow \text{ult}} &= 1.5500, \\ \text{Undeveloped proportion} &= 1 - \frac{1}{1.5500} \approx 0.3548 \quad (35.48\%), \\ \text{Developed proportion} &= \frac{1}{1.5500} \approx 0.6452 \quad (64.52\%).\end{aligned}$$

The decomposition into the observed and a priori components is:

$$\begin{aligned}\text{Ultimate}_{\text{BF}} &= \underbrace{1,050}_{\text{observed component}} + \underbrace{1,580 \times \left(1 - \frac{1}{1.5500}\right)}_{\text{a priori component}}, \\ &\approx 1,050 + 560.6 \\ &= 1,610.6 \text{ COP million.}\end{aligned}$$

Therefore,

$$\text{IBNR}_{\text{BF}} \approx 1,610.6 - 1,050 = 560.6 \text{ COP million.}$$

$\text{IBNR}_{\text{BF}}(2024) \approx 560.6 \text{ COP million}$

Comparison of methods for 2024

Method	Ultimate	IBNR
Chain-Ladder	1,627.5	577.5
Bornhuetter-Ferguson	1,610.6	560.6

Amounts are expressed in COP millions. The difference is:

$$\Delta \approx 577.5 - 560.6 = 16.9 \text{ COP million} \quad (\text{BF is more conservative}).$$

The difference reflects the fact that the a priori estimate (1,580) is lower than the CL-projected ultimate (1,627.5). BF acts as a weighted average: 64.52% of the weight is assigned to observed data (claim experience) and 35.48% to the a priori estimate. With so little development observed in 2024, BF moderates the optimism of the a priori estimate against a triangle that could be distorted by sampling fluctuations.

Question 04

Methodological judgment

In which of the following situations should a senior actuary prefer the Bornhuetter-Ferguson method over Chain-Ladder?

- A. The triangle contains 15 years of development, the factors are stable from year to year, and the oldest years are 97% developed.
- B. The most recent accident year is being reserved, with only one observed development period representing just 42% of expected ultimate claims.
- C. The line has a tail factor of exactly 1.0, and all claims are reported within the same year in which they occur.
- D. The company has just launched the line and has no a priori estimate of the expected loss ratio.

Answer

Option B

BF is superior when the accident year has **low statistical credibility**, meaning that the paid proportion is small relative to the expected ultimate. With a single development period representing 42% of the ultimate, the company's own data are insufficient for the CL factor to be reliable — an atypical claim on that diagonal can significantly bias the result.

Option A describes the ideal scenario for pure Chain-Ladder (a mature triangle and stable factors). Option C (tail = 1, immediate reporting) indicates a short-tail line where Chain-Ladder works well. Option D rules out BF because the method necessarily requires an ELR to work — without an a priori estimate, BF is not applicable.

***Rule of thumb:** the implicit weight of the a priori estimate in BF is $1 - 1/\text{CDF}$. When this weight exceeds 60%, many senior actuaries supplement it with the Cape Cod method or a BF/CL blend to reduce sensitivity to the ELR.*

Question 05

Multiple choice · Colombian regulation

Under Decree 2555 of 2010 and the current regulations of the Financial Superintendence of Colombia, which technical reserve is designed to cover claims that have already occurred but have not been settled or paid, including IBNR?

- A. Unexpired Risk Reserve (RRC): calculated as the unearned portion of the technical premium at the end of the period.
- B. Outstanding Claims Reserve (RSP): includes reported claims being processed, claims under litigation, and incurred but not reported claims (IBNR).
- C. Mathematical Reserve: specific to life insurance, calculated as the actuarial present value of future obligations.
- D. Claims Fluctuation Reserve: established to absorb statistical fluctuations relative to expected claims experience.

Answer

Option B

Decree 2555 of 2010 (Book 31, Part 2, Articles 31.2.3.1 and following) establishes the mandatory technical reserves for insurers in Colombia:

RSP — Outstanding Claims Reserve: covers obligations arising from claims that occurred before the valuation date and have not been settled or paid. It comprises three subcategories: (i) reported claims undergoing adjustment and settlement, (ii) claims subject to litigation or arbitration, and (iii) *incurred but not reported claims (IBNR)*, which must be estimated using documented actuarial methods approved by the SFC.

RRC: unearned premium reserve — covers the future risk of in-force policies, not past claims.

Mathematical Reserve: exclusive to long-term life insurance, calculated as the actuarial present value of future benefits less the present value of future premiums.

Fluctuation Reserve: a prospective statistical buffer, not directly linked to individual claims or IBNR.

Question 06

Short answer · Advanced concepts

Explain what the *tail factor* is in the Chain-Ladder method. For which types of insurance lines is it critical to estimate it correctly, and why does underestimating it constitute the actuarial risk with the greatest impact on long-term reserves?

Answer

The **tail factor**, f_{tail} , is the multiplier that projects cumulative claims from the last development period observable in the triangle to their *final ultimate value*:

$$\text{Ultimate} = \text{Latest diagonal} \times \text{CDF}_{\text{observed}} \times f_{\text{tail}}.$$

Where:

- $f_{\text{tail}} = 1.0$ implies that the triangle captures the full development.
- $f_{\text{tail}} > 1.0$ implies late claims not observed in the triangle.

Lines where it is critical

- **Liability insurance:** lawsuits may take 10–25 years to reach a final resolution.
- **Occupational risks (ARL) / workers' compensation:** disability and survivor pensions involve lifelong payments.
- **Occupational diseases:** the latency between exposure and diagnosis (asbestosis, silicosis, noise-related diseases) may exceed 20 years.
- **Medical liability:** claims for long-term harm to patients.

Risk of underestimation

If f_{tail} is prematurely set to 1.0, the projected ultimate is artificially low. In long-tail lines, the error in the tail factor may represent 30–50% of total IBNR — exceeding the sum of all errors in intermediate factors. The impact is delayed: reserves appear adequate for years until late claims mature and create a shortfall requiring abrupt reserve strengthening, with a direct impact on the underwriting result and solvency.

Question 07

Expert judgment

In the context of Bornhuetter-Ferguson, define the *Expected Loss Ratio* (ELR) and describe at least three sources of information that a senior actuary can use to estimate it when the company has little historical experience of its own in the line. What risks does a miscalibrated ELR introduce in either direction?

Answer

The **Expected Loss Ratio (ELR)** is the a priori estimate of the ultimate loss ratio relative to earned premium:

$$\text{ELR} = \frac{\text{Expected ultimate claims}}{\text{Earned premium}}.$$

In BF:

$$\begin{aligned}\text{Ultimate} &= \text{Paid} + \text{ELR} \times \text{Earned premium} \times \left(1 - \frac{1}{\text{CDF}}\right), \\ \text{IBNR} &= \text{ELR} \times \text{Earned premium} \times \left(1 - \frac{1}{\text{CDF}}\right).\end{aligned}$$

Sources for estimating the ELR with limited internal history

1. **Mature years from the same portfolio:** calculate the ultimate loss ratios (ultimate LR) of the older, fully developed cohorts in the triangle and project them, adjusting for inflation, changes in business mix, and exposure.
2. **Market statistics — FASECOLDA / Swiss Re Sigma:** in Colombia, FASECOLDA publishes loss ratio statistics by line of business with several years of history. For lines with limited internal history, the market benchmark is the most objective and auditable technical reference.
3. **ELR implicit in technical pricing:** the risk premium calculated in the pricing model embeds an expected loss ratio. If pricing was rigorous (and there was no significant adverse selection), the ELR from the rating model is the source most consistent with the business assumptions — it avoids the circularity of using the triangle for projection and then using the same triangle for calibration.
4. **Benchmarking with group reinsurers:** in HDI's case, Hannover Re provides portfolio analyses during negotiations for XL and quota share contracts. Its global view of the line by region is a high-quality expert source for calibrating the ELR when the company's own portfolio is new.

Risks of a miscalibrated ELR

- **Overestimated ELR:** excessive reserves → the underwriting result is artificially depressed, ROE appears lower, and this may lead to incorrect pricing decisions or withdrawal from the line.
- **Underestimated ELR:** insufficient reserves → the risk materializes later, when claims mature and exceed the booked reserve; this requires reserve strengthening in future periods, directly affecting the income statement and solvency ratios — the scenario with the greatest systemic risk.